

DevOps Engineer

Step 3: Learn Docker

CareerCompass — Detailed Learning Notes

What this step is about

Understand containerization well enough to package and run real applications.

Core Concepts

- Learn images, containers, Dockerfiles, registries, volumes, networks and Docker Compose.
- Understand that containers are isolated processes sharing the host kernel; they are not full virtual machines.
- Use small base images, pinned versions, non-root users where appropriate and external secret management.
- Learn container logs, health checks and resource considerations.

Skills You Should Be Able To Demonstrate

- Write and build a Dockerfile.
- Tag and run images.
- Use volumes and networks.
- Create a multi-container Compose stack.
- Inspect logs and container metadata.

Hands-on Project / Practice

- Containerize a Flask/Node app plus database and reverse proxy.
- Use Compose for local orchestration and environment variables for configuration.

Recommended References

- Docker Documentation — <https://docs.docker.com/>
- Docker Tutorial for Beginners — <https://www.youtube.com/watch?v=fqMOX6JJhGo>
- Docker in 2025 — <https://www.youtube.com/watch?v=eFgoK4GRqiw>